

to job losses or complete occupational obsolescence. Rather, it indicates the proportion of occupations that may undergo substantial changes due to AI integration. The impact on employment will likely be more complex, involving occupational transformations, the emergence of new roles, and potential displacement in some sectors.

Our analysis shows that in the United States and Mexico, women are more vulnerable to AI displacement, partly due to their significant presence in office and administrative jobs. In the United States, low-educated and low-wage workers are particularly exposed, while in Mexico, both low- and medium-skilled workers, especially those in middle-income formal jobs, are more vulnerable, whereas informal jobs remain less exposed. This suggests that AI adoption could exacerbate income inequalities by disproportionately impacting lower- and middle-income workers, echoing, to some extent, findings from previous technological advancements (Autor *et al.*, 2006; Autor and Dorn, 2013; Autor, 2019; Acemoglu and Restrepo, 2022).

To better understand the rationale behind the synthetic AI survey assessments, we conducted follow-up surveys, asking the model to explain the considerations behind each occupation’s estimation. These responses revealed that the model’s rationale focuses on several key factors: current and expected technological capabilities and limitations, the routine and non-routine nature of tasks, ethical considerations, the regulatory environment, and cost-benefit analyses. The models evaluated AI’s current abilities and future potential, emphasized the likelihood of replacing routine tasks, and frequently mentioned ethical considerations and safety concerns, particularly for roles with direct human interaction. Regulatory factors were noted for industries with strict standards, and cost-benefit analyses influenced the economic feasibility of automation.

We conducted several alternative exercises to examine how the GENOE index changes with different methodologies and prompts. These variations included omitting occupation names, excluding ethical considerations, assessing tasks individually, and concentrating on specific AI technologies. Most notably, when we instructed the model to disregard ethical, regulatory, and social considerations and focus purely on AI’s technical abilities, we observed a significant increase in the estimated exposure across occupations. This alternative GENOE index, which we term GENOE-Tech, provides an estimation of AI’s potential impact based solely on its growing technical capabilities.

This paper makes several contributions to the literature on AI’s impact on employment. First, it overcomes some limitations of the existing indexes that measure occupational exposure to AI (Brynjolfsson *et al.*, 2018; Webb, 2019; Felten *et al.*, 2021; Tolan *et al.*, 2021; Pizzinelli *et al.*, 2023; Eloundou *et al.*, 2024). Most existing indexes disaggregate occupations into collections of separate tasks, skills, or abilities. The first restriction of this approach is that it overlooks more complex interrelations between tasks or abilities. For instance, journalists and technical writers share several automatable routine tasks related to writing, editing, and generating content. Yet, routine roles in journalism often interact with non-routine roles like creating engaging narratives and storytelling, evaluating the significance of news, and engaging in investigative reporting, all of which require

human judgment, critical thinking, and persuasion. This interrelation will result in a lower AI exposure of journalism routine tasks than those of technical writers.

Another limitation of this approach is that it does not account for the ethical and social context of different occupations performing similar types of tasks. Consider, for example, judges and credit analysts. Both make decisions based on evaluating extensive information, and both roles can significantly impact individuals' lives. However, the nature and scope of their impact differ in important ways. Judges deliver verdicts that can dramatically affect individuals' well-being, freedom, rights, and long-term societal standing. Their role demands a profound understanding of ethical principles and the ability to balance complex cases. The ethical weight and social impact of their decisions, in addition to requiring human judgment and empathy, make it unlikely that societies would accept these roles being legitimately replaced by AI. Credit analysts, while also impacting individual well-being through their decisions on loan approvals or credit ratings, typically operate within a more narrowly defined financial context. Their decisions can certainly affect a person's economic opportunities and quality of life, but generally do not carry the same weight of personal liberty or broader societal impact as judicial decisions. This comparison highlights the importance of considering not just the technical similarities in task types, but also the broader ethical, social, and institutional contexts in which different occupations operate when assessing the potential for AI integration or replacement.

Our approach corrects the limitations of previous indexes by integrating a more holistic task-based analysis that captures the interrelations and complementarities between tasks within occupations. Unlike previous methods that disaggregate occupations into isolated tasks, skills, or abilities, our AI-generated index considers the entire vector of tasks, reflecting a complete view of occupational contributions in production. To address the second limitation, we provide the model in the surveys with both the name of the occupation and its vector of tasks, allowing it to estimate the overall likelihood of AI replacement without imposing restrictive assumptions. This method allows the model to consider the ethical and social context of each occupation organically rather than through predefined criteria, offering a more holistic evaluation of exposure and replacement. Alternative prompt specifications and follow-up surveys suggest that our method effectively accounts for these considerations.

This second concern is also raised by Pizzinelli *et al.* (2023) and is somewhat addressed by their complementarity-adjusted occupational exposure index (C-AIOE). This measure incorporates factors such as work contexts and job zones to account for the likelihood that AI will complement rather than replace human labor.⁴ Our approach goes further by imposing fewer assumptions

⁴Notwithstanding its advancements over previous indexes, the work contexts and job zones retrieved from the O*NET contain several limitations and assumptions. Firstly, it assumes that higher levels of education and training automatically correlate with better integration of AI, which may not account for the specific skill sets and practical contexts required in each occupation. Additionally, the chosen contexts, such as face-to-face communication, responsibility for others' health, and physical proximity, might not fully capture the ethical reasoning required in certain roles, nor the degree of public resistance to AI replacing certain decision-making roles. Furthermore, as the index is built upon the AIOE, it relies solely on the isolated similarity of abilities to AI developments and thus does not account for the exposure of tasks and their interrelations in the workplace.